

11. If A is square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to
(A) A (B) $I - A$ (C) I (D) $3A$

$$(I + A)^3 - 7A$$

$$I^3 + A^3 + 3I^2A + 3IA^2 - 7A$$

$$I + A \cdot A^2 + 3IA + 3A^2 - 7A$$

$$I + A \cdot A + 3A + 3A - 7A$$

$$I + A^2 + 6A - 7A$$

$$I + A - A$$

$$I$$

$$IA = A$$

$$OA = O$$

$$I^2 = I^3 = I^4 = I$$



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$$A = [a]$$

$$|A| = |a| = a$$

$$|A| = a_{11}a_{22} - a_{21}a_{12}$$

REMARKS

- (i) For matrix A, $|A|$ is read as determinant of A and not modulus of A.
- (ii) Only square matrices have determinants.

4.2.1 Determinant of a matrix of order one

Let $A = [a]$ be the matrix of order 1, then determinant of A is defined to be equal to a

4.2.2 Determinant of a matrix of order two

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a matrix of order 2×2 ,

then the determinant of A is defined as:

$$\det(A) = |A| = \Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

Example 1 Evaluate $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix}$ = $2(2) - 4(-1) = 4 + 4 = 8$

Solution We have $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = 2(2) - 4(-1) = 4 + 4 = 8.$

Example 2 Evaluate $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$ = $x(x) - (x-1)(x+1) = x^2 - (x^2 - 1)$

Solution We have

$$\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix} = x(x) - (x+1)(x-1) = x^2 - (x^2 - 1) = x^2 - x^2 + 1 = 1$$

$$= x^2 - x^2 + 1 = 1$$

4.2.3 Determinant of a matrix of order 3×3

Determinant of a matrix of order three can be determined by expressing it in terms of second order determinants. This is known as expansion of a determinant along a row (or a column). There are six ways of expanding a determinant of order

Example 3 Evaluate the determinant

$$\Delta \Rightarrow \begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

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Expand Δ along R_1

$$\Delta = (+1)(\underline{3 \times 0} - \underline{1 \times 0}) - 2(-\underline{1 \times 0} - 4 \times 0) + 4(-1 \times 1 - 4 \times 3)$$

Expand along R_2

$$\Delta = -(-1)(2 \times 0 - 1 \times 4) + 3(1 \times 0 - 4 \times 4) - 0(1 \times 1 - 4 \times 4)$$

Expand along C_1

$$\Delta = (+1)(3 \times 0 - 1 \times 0) - (-1)(2 \times 0 - 1 \times 4) + 4(2 \times 0 - 3 \times 4)$$

EXERCISE 4.1

Evaluate the determinants in Exercises 1 and 2.

1. $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$

2. (i) $\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

(ii) $\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$

$$= \underbrace{(x^2 - x + 1)(x + 1)} - (x + 1)(x - 1)$$
$$= (x^3 + 1) - (x^2 - 1)$$
$$= x^3 - x^2 + 2$$

3. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

4. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then show that $|3A| = 27|A|$

5. Evaluate the determinants

(i) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

(ii) $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

EXERCISE 4.1

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1. $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$

2. (i) $\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

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3. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

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(i) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

(ii) $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

Note:- (imp)

$A \rightarrow$ order n

$|kA| = k^n |A|$

order 3

$|7A| = 7^3 |A|$

order 2

$|2A| = 2^2 |A|$

$|9A| = 9^2 |A|$

$|7A| = 7^2 |A|$



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Q7C)

$$2 \times 1 - 4 \times 5 = 2x^2 - 6 \times 4$$

$$2 - 20 = 2x^2 - 24$$

$$-18 = 2x^2 - 24$$

$$6 = 2x^2$$

$$x^2 = 3$$

$$x = \pm \sqrt{3}$$

$$(iii) \begin{vmatrix} -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$$

$$(iv) \begin{vmatrix} 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

6. If $A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}$, find $|A|$

7. Find values of x , if

$$(i) \begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$$

$$(ii) \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$$

8. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to

- (A) 6 (B) ± 6 (C) -6 (D) 0

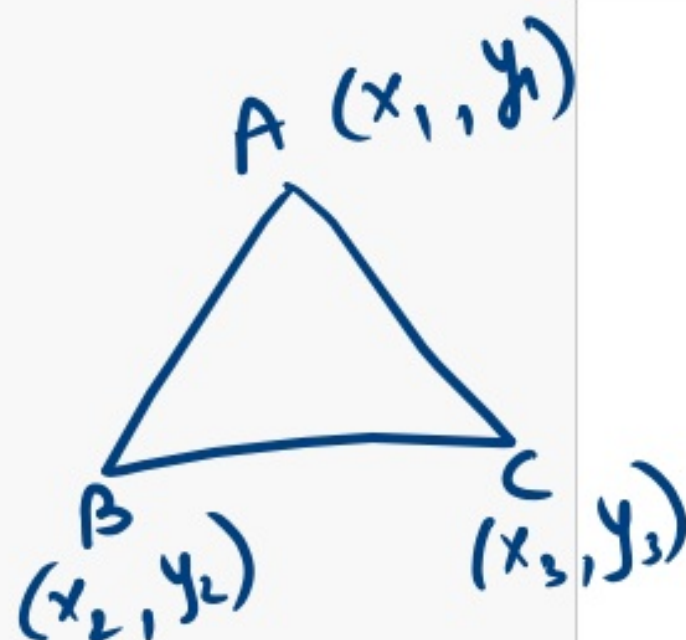
4.3 Area of a Triangle

In earlier classes, we have studied that the area of a triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , is given by the expression $\frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$. Now this expression can be written in the form of a determinant as

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \dots (1)$$

Remarks

- (i) Since area is a positive quantity, we always take the absolute value of the determinant in (1).
- (ii) If area is given, use both positive and negative values of the determinant for calculation.
- (iii) The area of the triangle formed by three collinear points is zero.



- (i) $\begin{vmatrix} 5 & 1 \\ 6 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$ (ii) $\begin{vmatrix} 4 & 5 \\ 2x & 5 \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ 2x & 5 \end{vmatrix}$
8. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to
- (A) 6 (B) ± 6 (C) -6 (D) 0

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Area of $\leftarrow \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \dots (1)$

Handwritten notes: "Area of" is written in blue. A red arrow points from the word "Area" to the determinant. A blue arrow points from the determinant to the word "modulus sign".

Remarks

- Since area is a positive quantity, we always take the absolute value of the determinant in (1).
- If area is given, use both positive and negative values of the determinant for calculation.
- The area of the triangle formed by three collinear points is zero.

Example 6 Find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$ and $(5, 1)$.

Solution The area of triangle is given by

$$\Delta = \frac{1}{2} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

Handwritten note: "det sign" in red.

Q3 (i)

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$4 = \frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$$

$$8 = \begin{vmatrix} -0 & +0 & -2(k-4) \end{vmatrix}$$

$$8 = -2(k-4)$$

$$8 = -2(k-4)$$

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3$$

This gives, $\frac{-3k}{2} = \pm 3$, i.e., $k = \mp 2$.

EXERCISE 4.2

- Find area of the triangle with vertices at the point given in each of the following :
 (i) (1, 0), (6, 0), (4, 3) (ii) (2, 7), (1, 1), (10, 8)
 (iii) (-2, -3), (3, 2), (-1, -8)
- Show that points
 A (a, b + c), B (b, c + a), C (c, a + b) are collinear.
- Find values of k if area of triangle is 4 sq. units and vertices are
 (i) (k, 0), (4, 0), (0, 2) (ii) (-2, 0), (0, 4), (0, k)
- (i) Find equation of line joining (1, 2) and (3, 6) using determinants.
 (ii) Find equation of line joining (3, 1) and (9, 3) using determinants.
- If area of triangle is 35 sq units with vertices (2, -6), (5, 4) and (k, 4). Then k is
 (A) 12 (B) -2 (C) -12, -2 (D) 12, -2

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$$2(k-4) = \pm 8$$

$$k-4 = \pm 4$$

$$k-4 = 4, -4$$

$$k = 12, 4$$

$A(1,3)$
 $B(0,0)$
 $P(x,y)$

$$= \frac{1}{2} [3(2-1) - 8(-4-5) + 1(-4-10)]$$

$$= \frac{1}{2} (3 + 72 - 14) = \frac{61}{2}$$

Example 7 Find the equation of the line joining A(1, 3) and B (0, 0) using determinants and find k if D(k, 0) is a point such that area of triangle ABD is 3sq units.

A, B, P → collinear

Solution Let P(x, y) be any point on AB. Then, area of triangle ABP is zero (Why?). So

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 3 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

Area of $\Delta ABP = 0$

This gives

$$\frac{1}{2}(y - 3x) = 0 \text{ or } y = 3x,$$

which is the equation of required line AB.

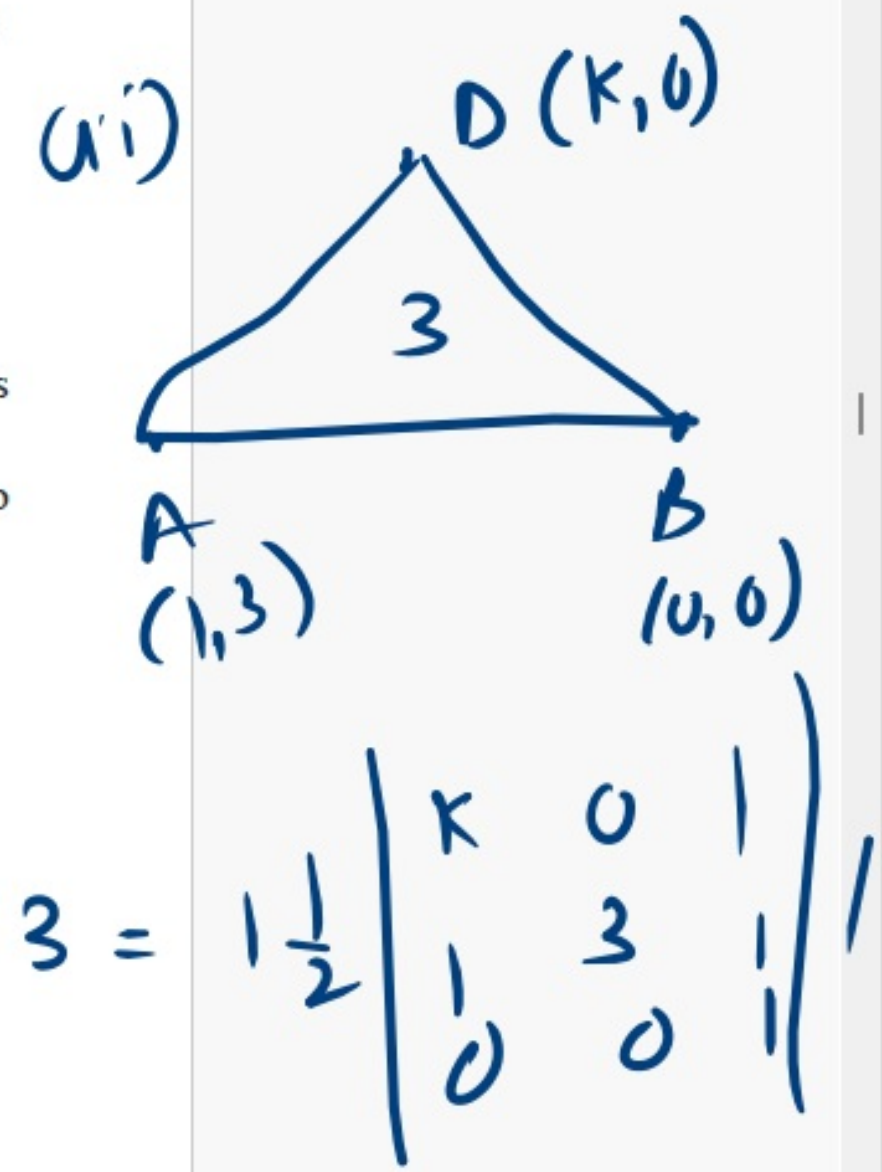
Also, since the area of the triangle ABD is 3 sq. units, we have

$$\frac{1}{2} \begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3$$

This gives, $\frac{-3k}{2} = \pm 3$, i.e., $k = \mp 2$.

EXERCISE 4.2

- Find area of the triangle with vertices at the point given in each of the following :
 - (1, 0), (6, 0), (4, 3)
 - (2, 7), (1, 1), (10, 8)
 - (-2, -3), (3, 2), (-1, -8)



$\frac{1}{2}(y - 3x) = 0$

 $y = 3x$

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MATHEMATICS

4.4 Minors and Cofactors

In this section, we will learn to write the expansion of a determinant in compact form using minors and cofactors.

Definition 1 Minor of an element a_{ij} of a determinant is the determinant obtained by deleting its i th row and j th column in which element a_{ij} lies. Minor of an element a_{ij} is denoted by M_{ij} .

Remark Minor of an element of a determinant of order n ($n \geq 2$) is a determinant of order $n - 1$.

Example 8 Find the minor of element 6 in the determinant $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & \underline{6} \\ 7 & 8 & 9 \end{vmatrix}$

Solution Since 6 lies in the second row and third column, its minor M_{23} is given by

$$M_{23} = \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 8 - 14 = -6 \text{ (obtained by deleting } R_2 \text{ and } C_3 \text{ in } \Delta).$$

Definition 2 Cofactor of an element a_{ij} , denoted by A_{ij} is defined by

$$A_{ij} = (-1)^{i+j} M_{ij}, \text{ where } M_{ij} \text{ is minor of } a_{ij}.$$

Example 9 Find minors and cofactors of all the elements of the determinant $\begin{vmatrix} 1 & -2 \\ 4 & 3 \end{vmatrix}$

Solution Minor of the element a_{ij} is M_{ij}

Here $a_{11} = 1$. So $M_{11} = \text{Minor of } a_{11} = 3$

$$\underline{\underline{\Sigma 98}}$$
$$M_{11} = 5 \times 9 - 8 \times 6$$
$$M_{12} = 4 \times 9 - 7 \times 6$$
$$M_{13} = 4 \times 8 - 5 \times 7$$
$$M_{23} = 1 \times 8 - 2 \times 7$$



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Eg 8

Cofactors = A_{ij} / C_{ij}

$$A_{11} = (-1)^{1+1} M_{11} = M_{11}$$

$$A_{12} = (-1)^{1+2} M_{12} = -M_{12}$$

$$A_{13} = (-1)^{1+3} M_{13} = M_{13}$$

$$A_{23} = (-1)^{2+3} M_{23} = -M_{23}$$

Definition 1 Minor of an element a_{ij} of a determinant is the determinant obtained by deleting its i th row and j th column in which element a_{ij} lies. Minor of an element a_{ij} is denoted by M_{ij} .

Remark Minor of an element of a determinant of order $n (n \geq 2)$ is a determinant of order $n - 1$.

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Definition 2 Cofactor of an element a_{ij} , denoted by A_{ij} is defined by $A_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is minor of a_{ij} .

Example 9 Find minors and cofactors of all the elements of the determinant $\begin{vmatrix} 1 & -2 \\ 4 & 3 \end{vmatrix}$

Solution Minor of the element a_{ij} is M_{ij}
 Here $a_{11} = 1$. So $M_{11} = \text{Minor of } a_{11} = 3$
 $M_{12} = \text{Minor of the element } a_{12} = 4$
 $M_{21} = \text{Minor of the element } a_{21} = -2$
 $M_{22} = \text{Minor of the element } a_{22} = 1$
 Now, cofactor of a_{ij} is A_{ij} . So
 $A_{11} = (-1)^{1+1} M_{11} = (-1)^2 (3) = 3$
 $A_{12} = (-1)^{1+2} M_{12} = (-1)^3 (4) = -4$

$$A_{11} = (-1)^{1+1} [5 \times 9 - 8 \times 6]$$

$$A_{23} = (-1)^{2+3} [1 \times 8 - 2 \times 7]$$

$$A_{31} = (-1)^{3+1} [2 \times 6 - 5 \times 3]$$

$$M_{11} = 3 \quad A_{11} = (-1)^{1+1} (3)$$

$$M_{12} = 4 \quad A_{12} = (-1)^{1+2} (4)$$

$$M_{21} = -2 \quad A_{21} = (-1)^{2+1} (-2)$$



EXERCISE 4.3

Write Minors and Cofactors of the elements of following determinants:

1. (i) $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$ (ii) $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

$A_{13} = (-1)^{1+3} M_{13} = 2 - y$
 $A_{23} = (-1)^{2+3} M_{23} = -(2 - x) = x - 2$
 $A_{33} = (-1)^{3+3} M_{33} = y - x$

Using Cofactors of elements of second row, evaluate $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$
 4. Using Cofactors of elements of third column, evaluate $\Delta = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$

5. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is Cofactors of a_{ij} , then value of Δ is given by
 (A) $a_{11} A_{31} + a_{12} A_{32} + a_{13} A_{33}$ (B) $a_{11} A_{11} + a_{12} A_{21} + a_{13} A_{31}$
 (C) $a_{21} A_{11} + a_{22} A_{12} + a_{23} A_{13}$ (D) $a_{11} A_{11} + a_{21} A_{21} + a_{31} A_{31}$

4.5 Adjoint and Inverse of a Matrix

In the previous chapter, we have studied inverse of a matrix. In this section, we shall discuss the condition for existence of inverse of a matrix.

To find inverse of a matrix A i.e. A^{-1} we shall first define adjoint of a matrix

eg:

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = \begin{matrix} R_1 \rightarrow \end{matrix} \underline{1} (5 \times 9 - 8 \times 6) - \underline{2} (4 \times 9 - 6 \times 7) + \underline{3} (4 \times 8 - 5 \times 7)$$

$\underline{a_{11}} \underline{A_{11}} + \underline{a_{12}} \underline{A_{12}} + \underline{a_{13}} \underline{A_{13}}$

$(-1)^{1+2} \quad (-1)^{1+3}$

U.V. up

Δ = sum of elements of a row/column multiplied with their respective cofactors

(R₂) $\Delta = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$

(R₃) $\Delta = a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33}$

(C₁) $\Delta = a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$

(C₃) $\Delta = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33}$

Note:-

If multiply elements of a row/column with the cofactors of another row/column = 0, Zero.

$$Q_{11} A_{21} + Q_{12} A_{22} + Q_{13} A_{23} = 0$$

→ Cofactors of 1st Column

$$Q_{21} A_{11} + Q_{22} A_{21} + Q_{23} A_{31} = 0$$

→ elements of 2nd Column

H.W

Ex 4.1 $\rightarrow 02, 04, 07, 08$

Example 7

Ex 4.2 $\rightarrow 02, 03, 04, 05$

Example 11

Ex 4.3 $\rightarrow 02(ii), 01(ii), 04, 05$

Q4

$$\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$\Delta = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33}$$

$$\Delta = (yz)(-1)^{1+3}(z-y) + zx(-1)^{2+3}(z-x) + xy(-1)^{3+3}(y-x)$$

$$\Delta = yz(2-y) - zx(2-x) + xy(y-x)$$

EXERCISE 4.3

Write Minors and Cofactors of the elements of following determinants:

$$(i) \begin{vmatrix} 2 & -4 \\ 1 & 0 \end{vmatrix}$$

$$(ii) \begin{vmatrix} a & c \\ b & d \end{vmatrix}$$

$$2. (i) \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{vmatrix}$$

3. Using Cofactors of elements of second row, evaluate $\Delta =$

$$\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

4. Using Cofactors of elements of third column, evaluate $\Delta =$

$$\begin{vmatrix} 1 & y & zx \\ 1 & z & xy \end{vmatrix}$$

5. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is Cofactors of a_{ij} , then value of Δ is given by

- (A) $a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33}$ (B) $a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31}$
 (C) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$ (D) $a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$

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In the previous chapter, we have studied inverse of a matrix. In this section, we shall discuss the condition for existence of inverse of a matrix.

To find inverse of a matrix A, i.e., A^{-1} we shall first define adjoint of a matrix

4.5.1 Adjoint of a matrix

Definition 2 The adjoint of a square matrix A = [a_{ij}] is defined as the transpose of

Q3

$$\Delta = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$$

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\text{adj } A = [\text{cofactor matrix of } A]$$

$$= \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix}'$$

$$\text{adj } A = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

sign changes

interchanged

Then $\text{adj } A = \text{Transpose of } \begin{bmatrix} A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$

Example 12 Find $\text{adj } A$ for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

Solution We have $A_{11} = 4, A_{12} = -1, A_{21} = -3, A_{22} = 2$

$$\text{adj } A = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

Remark For a square matrix of order 2, given by

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

The $\text{adj } A$ can also be obtained by interchanging a_{11} and a_{22} and by changing signs of a_{12} and a_{21} , i.e.,

$$\text{adj } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \xrightarrow[\text{Change sign}]{\text{Interchange}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

We state the following theorem without proof.

Theorem 1 If A be any given square matrix of order n , then

$$A(\text{adj } A) = (\text{adj } A)A = |A|I,$$

where I is the identity matrix of order n
Verification

cofactor of $a_{12} = 3 = (-1)^{1+2} (1) = -1$

Note:- Trick works to find $\text{adj } A$ is only order 2 matrix.

or $A^{-1} = 4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$

Hence $A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$

EXERCISE 4.4

Find adjoint of each of the matrices in Exercises 1 and 2.

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
2. $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$

Verify $A(\text{adj } A) = (\text{adj } A)A = |A|I$ in Exercises 3 and 4

3. $\begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$
4. $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$

Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

5. $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$
6. $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$
7. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

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$$A(\text{adj } A) = (\text{adj } A)A = |A|I$$

$$|A|I = 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Q3

$$\text{adj } A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

$$A(\text{adj } A) = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(\text{adj } A)A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$|A| = -12 - (-12) = 0$$

Similarly, we can show $(adj A) A = |A| I$

Hence $A (adj A) = (adj A) A = |A| I$

Definition 4 A square matrix A is said to be singular if $|A| = 0$.

For example, the determinant of matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$ is zero

Hence A is a singular matrix.

Definition 5 A square matrix A is said to be non-singular if $|A| \neq 0$

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Then $|A| = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = -2 \neq 0$.

Hence A is a nonsingular matrix

We state the following theorems without proof.

Theorem 2 If A and B are nonsingular matrices of the same order, then AB and BA are also nonsingular matrices of the same order.

Theorem 3 The determinant of the product of matrices is equal to product of their respective determinants, that is, $|AB| = |A| |B|$, where A and B are square matrices of the same order

Remark We know that $(adj A) A = |A| I = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}, |A| \neq 0$

Writing determinants of matrices on both sides, we have

$$|(adj A) A| = \begin{vmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{vmatrix}$$

eg:-

$$A = \begin{bmatrix} 1 & 2 & x \\ 3 & 5 & 8 \\ x & 7 & 0 \end{bmatrix}$$

$A \rightarrow$ singular matrix
find x

$$|A| = 0$$

$$1(0 - 56) - 2(-8x) + x(21 - 5x) = 0$$

Q8

$$\text{adj } A = \begin{bmatrix} -3 & +3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$$

5. $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$

6. $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 6 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$$

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$$\text{adj } A = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

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Q10

$$|A| = 1(-3-0) = -3$$

8. $\begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$

9. $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$

10. $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$

11. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$

12. Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$. Verify that $(AB)^{-1} = B^{-1} A^{-1}$.

13. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1} .

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \frac{1}{-3} \begin{bmatrix} -3 & 0 & 6 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$



eg 15

Proof

$$A^2 - 4A + I = O$$

Post multiply this equation with A^{-1}

$$A^2 A^{-1} - 4A A^{-1} + I A^{-1} = O A^{-1}$$

$$A \cdot A A^{-1} - 4I + A^{-1} = O$$

$$A \cdot I - 4I + A^{-1} = O$$

$$A - 4I + A^{-1} = O$$

Example 15 Show that the matrix $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ satisfies the equation $A^2 - 4A + I = O$, where I is 2×2 identity matrix and O is 2×2 zero matrix. Using this equation, find A^{-1} .

Solution We have $A^2 = A \cdot A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$

Hence $A^2 - 4A + I = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

Now $A^2 - 4A + I = O$

Therefore

$$A A - 4A = -I$$

or

$$A A (A^{-1}) - 4 A A^{-1} = -I A^{-1} \text{ (Post multiplying by } A^{-1} \text{ because } |A| \neq 0)$$

or

$$A (A A^{-1}) - 4I = -A^{-1}$$

or

$$AI - 4I = -A^{-1}$$

or

$$A^{-1} = 4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

EXERCISE 4.4

Find adjoint of each of the matrices in Exercises 1 and 2.

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

2. $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$

Verify $A (\text{adj } A) = (\text{adj } A) A = |A| I$ in Exercises 3 and 4

$$AB = BA = I$$

$$B = A^{-1}$$

$$A A^{-1} = A^{-1} A = I$$

$$A^{-1} = 4I - A$$

$$A^{-1} = 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

Q15

$A^3 - 6A^2 + 5A + 11I = 0$

Post multiply A^{-1}

$A^3 A^{-1} - 6A^2 A^{-1} + 5A A^{-1} + 11I A^{-1} = 0 A^{-1}$

$A^2 I - 6A I + 5I + 11A^{-1} = 0$

$11A^{-1} = 6A - A^2 - 5I$

$\begin{bmatrix} 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$

12. Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$. Verify that $(AB)^{-1} = B^{-1} A^{-1}$.

13. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1} .

14. For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$.

15. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ -1 & -1 & 3 \end{bmatrix}$
Show that $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1} .

16. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$
Verify that $A^3 - 6A^2 + 9A - 4I = O$ and hence find A^{-1} .

17. Let A be a nonsingular square matrix of order 3×3 . Then $|adj A|$ is equal to
(A) $|A|$ (B) $|A|^2$ (C) $|A|^3$ (D) $3|A|$

18. If A is an invertible matrix of order 2, then $\det(A^{-1})$ is equal to
(A) $\det(A)$ (B) $\frac{1}{\det(A)}$ (C) 1 (D) 0

4.6 Applications of Determinants and Matrices

In this section, we shall discuss application of determinants and matrices for solving the system of linear equations in two or three variables and for checking the consistency of the system of linear equations.

Q17

$A \Rightarrow$ order 3

$$\underline{\underline{|adj A| = |A|^{n-1}}}$$

$$= |A|^{3-1} = |A|^2$$

Q18

$$\det(A^{-1}) = |A^{-1}|$$

$$|A^{-1}| = \left| \frac{1}{|A|} adj A \right|$$

$$\begin{bmatrix} 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$$

12. Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$. Verify that $(AB)^{-1} = B^{-1} A^{-1}$.

13. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1} .

14. For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$.

15. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$

Show that $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1} .

16. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

Verify that $A^3 - 6A^2 + 9A - 4I = O$ and hence find A^{-1}

17. Let A be a nonsingular square matrix of order 3×3 . Then $|adj A|$ is equal to
(A) $|A|$ (B) $|A|^2$ (C) $|A|^3$ (D) $3|A|$

18. If A is an invertible matrix of order 2, then $\det(A^{-1})$ is equal to
(A) $\det(A)$ (B) $\frac{1}{\det(A)}$ (C) 1 (D) 0

4.6 Applications of Determinants and Matrices

In this section, we shall discuss application of determinants and matrices for solving the system of linear equations in two or three variables and for checking the consistency of the system of linear equations

let $\frac{1}{|A|} = K$ for simplified

$$|A^{-1}| = |K adj A|$$

use formula

$$|KB| = K^n |B|$$

$$|A^{-1}| = K^2 |adj A|$$

$$|A^{-1}| = K^2 |A|^{2-1}$$

$$|A^{-1}| = \frac{1}{|A|^2} \cdot |A| = \frac{1}{|A|}$$

Q18

$$\det(A^{-1}) = |A^{-1}| = \frac{1}{|A|} = \frac{1}{\det(A)}$$

↳ only for this question (when order = 2)

$$|A^{-1}| = \frac{1}{|A|} \rightarrow \text{not always true.}$$

H.W

Exercise 4.4 $\rightarrow$ All questions

Example -15, 14, 13

Application of matrices & Determinants

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$AX = B$$

$$a_1x + b_1y = d_1$$
$$a_2x + b_2y = d_2$$

→ unique solⁿ

→ infinite many solⁿ

→ no solution

$$AX = B$$

$$A^{-1}AX = A^{-1}B$$

$$\left[X = \frac{1}{|A|} (\text{adj } A) B \right]$$

Case I

$|A| \neq 0$; unique solution

Case II

$|A| = 0$; $A \rightarrow$ singular matrix

and $(\text{adj } A) B = 0$ then infinite many solⁿ

Case III

$|A| = 0$ but $(\text{adj } A) B \neq 0$ then no solution

Consistent solutions

inconsistent solution

If $(adj A) B = O$, then system may be either consistent or inconsistent according as the system have either infinitely many solutions or no solution.

Example 16 Solve the system of equations

$$\begin{aligned} 2x + 5y &= 1 \\ 3x + 2y &= 7 \end{aligned}$$

Handwritten: $\begin{cases} 2x + 5y = 1 \\ 3x + 2y = 7 \end{cases} \Rightarrow \begin{cases} 2x + 5y - 1 = 0 \\ 3x + 2y - 7 = 0 \end{cases}$

Solution The system of equations can be written in the form $AX = B$, where

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

Now, $|A| = -11 \neq 0$, Hence, A is nonsingular matrix and so has a unique solution.

Note that

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

Therefore

$$X = A^{-1}B = -\frac{1}{11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

i.e.

$$\begin{bmatrix} x \\ y \end{bmatrix} = -\frac{1}{11} \begin{bmatrix} -33 \\ 11 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \checkmark$$

Hence

$$x = 3, y = -1$$

Example 17 Solve the following system of equations by matrix method.

$$\begin{aligned} 3x - 2y + 3z &= 8 \\ 2x + y - z &= 1 \\ 4x - 3y + 2z &= 4 \end{aligned}$$

Solution The system of equations can be written in the form $AX = B$, where

Handwritten: $AX = B$

Handwritten: $\Rightarrow X = A^{-1}B$

Handwritten: $X = \frac{1}{|A|} (adj A) B$

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15. If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} solve the system of equations

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

— A; X; B

16. The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹ 60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹ 90. The cost of 6 kg onion 2 kg wheat and 3 kg rice is ₹ 70. Find cost of each item per kg by matrix method.

Miscellaneous Examples

Example 19 Use product $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 9 & 2 & 3 \\ 6 & 1 & 2 \end{bmatrix}$ to solve the system of equations

$$x - y + 2z = 1$$

$$2y - 3z = 1$$

$$3x - 2y + 4z = 2$$

Q15

$$A^{-1} = \frac{1}{|A|} (\text{adj } A) = \text{final answer}$$

$$AX = B$$

$$X = A^{-1} B$$

$$X = (\text{final ans}) \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$



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Ex 18
 Let three numbers
 be x, y, z
 $x + y + z = 6$
 $x + y + 3z = 11$
 $x + z = 2y$
 $\hookrightarrow x - 2y + z = 0$

i.e.
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -17 \\ -34 \\ -51 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Hence $x = 1, y = 2$ and $z = 3$.

Example 18 The sum of three numbers is 6. If we multiply third number by 3 and add second number to it, we get 11. By adding first and third numbers, we get double of the second number. Represent it algebraically and find the numbers using matrix method.

Solution Let first, second and third numbers be denoted by x, y and z , respectively. Then, according to given conditions, we have

$$\begin{aligned} x + y + z &= 6 \\ y + 3z &= 11 \\ x + z &= 2y \text{ or } x - 2y + z = 0 \end{aligned}$$

This system can be written as $A X = B$, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & 2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Here $|A| = 1(1+6) - (0-3) + (0-1) = 9 \neq 0$. Now we find $\text{adj } A$

$$\begin{aligned} A_{11} &= 1(1+6) = 7, & A_{12} &= -(0-3) = 3, & A_{13} &= -1 \\ A_{21} &= -(1+2) = -3, & A_{22} &= 0, & A_{23} &= -(-2-1) = 3 \\ A_{31} &= (3-1) = 2, & A_{32} &= -(3-0) = -3, & A_{33} &= (1-0) = 1 \end{aligned}$$

$$\text{adj } A = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Hence

Q16

Let cost of onion is ₹ x
 " of wheat is ₹ y
 " of rice is ₹ z

$$4x + 3y + 2z = 60$$

$$2x + 4y + 6z = 90$$

$$6x + 2y + 3z = 70$$

$$\begin{bmatrix} 1 & 1 & -2 \end{bmatrix}$$

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹ 60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹ 90. The cost of 6 kg onion 2 kg wheat and 3 kg rice is ₹ 70. Find cost of each item per kg by matrix method.

Miscellaneous Examples

Example 19 Use product $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 9 & 2 & 3 \\ 6 & 1 & 2 \end{bmatrix}$ to solve the system of equations

$$x - y + 2z = 1$$

$$2y - 3z = 1$$

$$3x - 2y + 4z = 2$$

Solution Consider the product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$

$$= \begin{bmatrix} -2 - 9 + 12 & 0 - 2 + 2 & 1 + 3 - 4 \\ 0 + 18 - 18 & 0 + 4 - 3 & 0 - 6 + 6 \\ -6 - 18 + 24 & 0 - 4 + 4 & 3 + 6 - 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$



16. The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹ 60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹ 90. The cost of 6 kg onion 2 kg wheat and 3 kg rice is ₹ 70. Find cost of each item per kg by matrix method.

Miscellaneous Examples

Example 19 Use product

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

$$\begin{aligned} x - y + 2z &= 1 \\ 2y - 3z &= 1 \\ 3x - 2y + 4z &= 2 \end{aligned}$$

Solution Consider the product

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix} \Rightarrow A^{-1}$$

$$= \begin{bmatrix} -2 - 9 + 12 & 0 - 2 + 2 & 1 + 3 - 4 \\ 0 + 18 - 18 & 0 + 4 - 3 & 0 - 6 + 6 \\ -6 - 18 + 24 & 0 - 4 + 4 & 3 + 6 - 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \checkmark$$

Hence

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

Now, given system of equations can be written, in matrix form, as follows

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Eg 19

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$AB = I$$

$$AB = BA = I$$

$$B = A^{-1}$$

$$\begin{aligned} \underline{A} \underline{A}^{-1} &= \underline{I} \\ \underline{A}^{-1} \underline{A} &= \underline{I} \end{aligned}$$

Q1
Expand it along R_1

$$\begin{aligned}
 & x(-x^2 - 1) - \sin \theta (-x \sin \theta - \cos \theta) + \cos \theta (-\sin \theta + x \cos \theta) \\
 & -x^3 - x + x \sin^2 \theta + \sin \theta \cos \theta - \sin \theta \cos \theta + x \cos^2 \theta \\
 & -x^3 - x + x(\sin^2 \theta + \cos^2 \theta) \\
 & -x^3 - x + x
 \end{aligned}$$

which is independent of θ
Hence proved

Hence

$$x = 0, y = 5 \text{ and } z = 3$$

Miscellaneous Exercises on Chapter 4

1. Prove that the determinant $\begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ is independent of θ .

2. Evaluate $\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$

3. If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$

4. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$. Verify that

(i) $[adj A]^{-1} = adj(A^{-1})$ (ii) $(A^{-1})^{-1} = A$

5. Evaluate $\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$

6. Evaluate $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x+y & x+y \end{vmatrix}$



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Q3

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$B^{-1} = \frac{1}{|B|} \text{adj } B$$

$$B^{-1}A^{-1} = \underline{A^{-1}B^{-1}}$$

1. Prove that the determinant $\begin{vmatrix} -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ is independent of θ .

2. Evaluate $\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$

3. If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$

4. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$. Verify that

(i) $[\text{adj } A]^{-1} = \text{adj } (A^{-1})$

(ii) $(A^{-1})^{-1} = A$

5. Evaluate $\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$

6. Evaluate $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$



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Qy L.H.S

$$(adj A) = [\quad]$$

$$(adj A)^{-1}$$

RHS

$$A^{-1} = [\quad]$$

$$adj(A^{-1}) = [\quad]$$

1. Prove that the determinant $\begin{vmatrix} -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ is independent of θ .

2. Evaluate $\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$

3. If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$

4. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$. Verify that

(i) $[adj A]^{-1} = adj (A^{-1})$ (ii) $(A^{-1})^{-1} = A$

5. Evaluate $\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$

6. Evaluate $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$

formula

Note:-

(i) $(adj A)^{-1} = adj(A^{-1})$

(ii) $(A^{-1})^{-1} = A$

Q7

Let

$$\frac{1}{x} = a$$

$$\frac{1}{y} = b$$

$$\frac{1}{z} = c$$

$$2a + 3b + 10c = 4$$

$$4a - 6b + 5c = 1$$

$$6a + 9b - 2c = 2$$

$$A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -2 \end{bmatrix}$$

100 MATHEMATICS

Using properties of determinants in Exercises 11 to 15, prove that:

7. Solve the system of equations

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$$

$$\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$$

Choose the correct answer in Exercise 17 to 19.

8. If x, y, z are nonzero real numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is

(A) $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

(B) $xyz \begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

(C) $\frac{1}{xyz} \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

(D) $\frac{1}{xyz} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$X = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow X = A^{-1}B$$

$$X = \frac{1}{|A|} (\text{adj } A) B$$

Using properties of determinants in Exercises 11 to 15, prove that:

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(D) $\frac{1}{xyz} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$|A| = 1200$$

$$X = \frac{1}{|A|} (\text{adj } A) B$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

$$a = \frac{1}{2}, b = \frac{1}{3}, c = \frac{1}{5}$$

$$\text{adj } A = \begin{bmatrix} 75 & 110 & 72 \\ 150 & -100 & 0 \\ 75 & 30 & -24 \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 30 & -24 \end{bmatrix}$$

$$(\text{adj } A) B = \begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$\frac{1}{x} = a = \frac{1}{2}$$

$$x = 2$$

$$\frac{1}{y} = b = \frac{1}{3}$$

$$y = 3$$

$$\frac{1}{z} = c = \frac{1}{5}$$

$$z = 5$$

verify

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{2}{2} + \frac{3}{3} + \frac{10}{5}$$

$$1 + 1 + 2$$

$$4 = 4$$

verified



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Q9

$$\det(A) = 2(1 + \sin^2 \theta)$$

$$0 \leq \theta \leq 2\pi$$

$$-1 \leq \sin \theta \leq 1$$

$$0 \leq \sin^2 \theta \leq 1$$

$$1 \leq 1 + \sin^2 \theta \leq 2$$

$$2 \leq 2(1 + \sin^2 \theta) \leq 4$$

$$2 \leq \det(A) \leq 4$$

Choose the correct answer in Exercise 17 to 19.

8. If x, y, z are nonzero real numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is

(A) $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

(B) $xyz \begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

(C) $\frac{1}{xyz} \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

(D) $\frac{1}{xyz} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

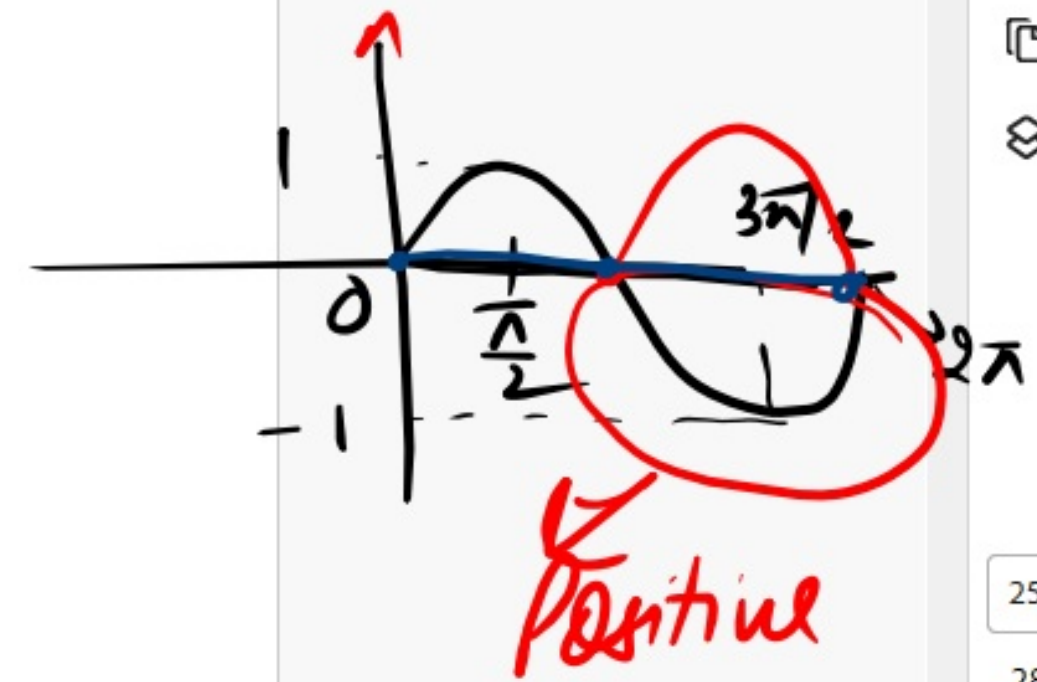
9. Let $A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$, where $0 \leq \theta \leq 2\pi$. Then

(A) $\det(A) = 0$

(B) $\det(A) \in (2, \infty)$

(C) $\det(A) \in (2, 4)$

☒ (D) $\det(A) \in [2, 4]$



H.W

Remaining questions of previous H.W

Raphael → 5

Emmanuel → 6

+

Example 9

+

Misc.

Exercise

Q1 to Q9

except Q4.